

Functions: The Personal Expense Tracker

Overview

In this assignment, you will develop a Personal Expense Tracker using Python. This application will allow users to input their expenses and keep track of their total spending. To get started, ensure that you download the “Personal_Expense_Tracker.py” template file attached to this assignment in Blackboard. Use the “Load” button in Mu to load the attached template file and start your work. Do note that double-clicking a python file will attempt to run the file. This can make it appear as though the file opened and closed quickly but did nothing. In fact, the program ran and closed in that small window. If you want to edit a file, you must load it into an editor as previously described. You should only make the necessary changes to the template based on the TODO comments.

Objectives

- To create a functional Personal Expense Tracker.
- To practice writing and using functions in Python.
- To handle user input and perform basic arithmetic operations.
- To reinforce concepts of error handling and user interaction in Python programming.

Requirements

- Your program should accurately track and display the total expenses.
- It must handle invalid inputs gracefully without crashing.
- The final output should display the total expenses after the user is done entering expenses.

Functions to Implement

1. **get_expense_amount**
 - Prompt the user to enter an expense amount or type 'done' to finish.
 - Handle non-numeric inputs and prompt the user again in such cases (except 'done').
 - Handle negative inputs and prompt the user again in such cases.
 - Return the entered amount or 'done' if the user finishes entering expenses.
 - Hint: remember that input() always returns a string.
2. **get_expense_description**
 - Prompt the user to enter a description for the expense.
 - Return the entered description.
3. **add_expense**
 - Add the entered expense amount to the total.
 - Return the new total.
4. **display_total**
 - Print the current total of expenses.
5. **main**
 - Coordinate the overall functionality of the program using the functions above.
 - Include a loop that continues to prompt for expenses until the user decides to finish.
 - Call the appropriate functions to get inputs, add expenses, and display totals.